

# RIFAT ERDEM SAHIN

## LLM Engineer | Large Language Model Specialist

---

\*\* Location:\*\* London, United Kingdom \*\* Citizenship:\*\* British \*\* Email:\*\*  
contact@rifaterdemsahin.com \*\* Phone:\*\* +44 7848 024173 \*\* LinkedIn:\*\*  
linkedin.com/in/rifaterdemsahin \*\* GitHub:\*\* github.com/rifaterdemsahin \*\*  
Portfolio:\*\* <https://rifaterdemsahin.com> \*\* Schedule a Call:\*\* <https://calendly.com/rifaterdem/schedule>

---

## PROFESSIONAL SUMMARY

Senior LLM Engineer specializing in **Large Language Models, fine-tuning, and production LLM systems**. Deep expertise in designing and implementing enterprise-scale LLM applications with focus on model optimization, inference scaling, and AI safety. Proven track record building LLM solutions that delivered 300% productivity improvements and 30% cost reductions across financial services, healthcare, and real estate sectors. Expert in transformer architectures and responsible AI deployment.

---

## CORE COMPETENCIES

**LLM Development & Fine-tuning** - Large Language Model fine-tuning with LoRA, QLoRA, and PEFT techniques - Custom model training for domain-specific applications - Model compression and quantization for production deployment - Prompt engineering and few-shot learning optimization

**LLM Infrastructure & Scaling** - Production LLM inference optimization and scaling - Model serving with TensorRT, vLLM, and TGI (Text Generation Inference) - GPU cluster management and resource optimization - Cost optimization strategies for LLM inference at scale

---

## KEY ACCOMPLISHMENTS

**2024 | Goldman Sachs | Muscat, Oman**

**Enterprise LLM Platform for Financial Services - Challenge:** Build scalable LLM platform for financial document analysis and automated compliance - **Solution:** Architected comprehensive LLM platform with fine-tuned models and automated inference scaling - **Impact:** - 300% improvement in document processing speed through optimized LLM inference - 30% reduction

in operational costs via efficient model serving - 95% accuracy in financial document classification and analysis - **Technologies:** Transformers, PyTorch, vLLM, CUDA, Kubernetes, MLflow

## 2023 | Ypsomed | Switzerland

**Medical LLM for Healthcare Documentation - Challenge:** Develop specialized LLM for medical device documentation and patient interaction - **Solution:** Fine-tuned domain-specific LLM with medical knowledge base integration - **Impact:** - 40% improvement in documentation accuracy through specialized model training - 35% reduction in manual documentation effort via LLM automation - Real-time medical insights enabling improved patient care - **Technologies:** Hugging Face, LoRA, Medical datasets, Azure OpenAI, RAG systems

## 2022 | Cushman & Wakefield | London, UK

**Real Estate LLM Analytics Platform - Challenge:** Build LLM system for property market analysis and automated reporting - **Solution:** Implemented fine-tuned LLM with real estate domain knowledge and market data integration - **Impact:** - 50% faster market analysis through automated LLM-powered insights - 90% improvement in report generation speed via natural language processing - Enhanced business intelligence enabling strategic property investment decisions - **Technologies:** GPT models, Fine-tuning frameworks, Real estate datasets, Azure ML

---

# TECHNICAL LLM EXPERTISE

## LLM Frameworks & Tools

**Models:** GPT-4, Claude, LLaMA, Mistral, BERT, T5, Custom transformer architectures **Fine-tuning:** LoRA, QLoRA, PEFT, Supervised Fine-Tuning (SFT), RLHF **Libraries:** Transformers, PyTorch, TensorFlow, Hugging Face, LangChain, LlamaIndex **Serving:** vLLM, TGI, TensorRT-LLM, Triton Inference Server, Ray Serve

## LLM Operations & Optimization

**Optimization:** Model quantization, Knowledge distillation, Pruning, TensorRT optimization **Scaling:** Distributed inference, Model parallelism, Pipeline parallelism, Load balancing **Monitoring:** Model performance tracking, Inference latency optimization, Cost monitoring **Safety:** Content filtering, Bias detection, Adversarial robustness, Alignment techniques

## Programming & Infrastructure

**Languages:** Python, CUDA, C++, Bash, SQL **ML Infrastructure:** Kubernetes, Docker, NVIDIA GPUs, MLOps pipelines **Cloud Platforms:** Azure OpenAI, AWS Bedrock, GCP Vertex AI, RunPod, Lambda Labs **Data Processing:** Pandas, NumPy, Apache Spark, Data preprocessing pipelines

---

## LLM SPECIALIZATIONS

### Custom Model Development

- Fine-tuned LLMs for financial, healthcare, and real estate domains
- Implemented RLHF (Reinforcement Learning from Human Feedback) pipelines
- Built custom tokenizers and preprocessing pipelines for specialized datasets
- Created model evaluation frameworks with domain-specific benchmarks

### Production LLM Systems

- Designed high-throughput LLM inference systems serving 1M+ daily requests
- Implemented adaptive batching and caching strategies reducing latency by 60%
- Built cost optimization systems reducing LLM inference costs by 50%
- Created automated model A/B testing and deployment pipelines

### LLM Safety & Alignment

- Implemented content filtering and safety guardrails for production LLM systems
  - Built bias detection and mitigation frameworks for responsible AI deployment
  - Created explainability tools for LLM decision making in regulated industries
  - Designed red-teaming frameworks for LLM robustness testing
- 

## PROFESSIONAL EXPERIENCE HIGHLIGHTS

**Senior LLM Engineer / AI Technical Lead** | 2020 - Present *Goldman Sachs, Ypsomed, Cushman & Wakefield*

- Led LLM initiatives across finance, healthcare, and real estate sectors
- Designed and implemented LLM systems supporting millions of daily interactions
- Established LLM best practices and production deployment frameworks

- Mentored engineering teams on transformer architectures and LLM optimization
- Delivered LLM solutions achieving 300% improvement in processing efficiency

**AI/ML Engineer** | 2016 - 2020 *Microsoft, Emerson, Various Fortune 500*

- Built machine learning solutions for digital transformation initiatives
- Led development of NLP and language model applications
- Established ML frameworks and deployment strategies
- Evangelized AI technologies through technical leadership and training

---

## EDUCATION

**Bachelor of Science** Southern New Hampshire University, USA | 2013

---

## CERTIFICATIONS & CONTINUOUS LEARNING

**AWS Certified Machine Learning Specialty Azure AI Engineer Associate NVIDIA Deep Learning Institute Certifications Certified Kubernetes Administrator (CKA)**

**Continuous Learning:** - Active contributor to LLM and transformer model open-source projects - Regular participant in AI research conferences and LLM community events - Following latest developments in transformer architectures and training techniques - Experimenting with cutting-edge LLM optimization and serving technologies

---

## THOUGHT LEADERSHIP & SPEAKING

- Technical blog on LLM engineering and optimization at rifaterdem-sahin.com
- Internal tech talks on LLM fine-tuning and production deployment
- Mentorship in AI/ML communities and LLM engineering teams
- LLM engineering examples and tutorials shared via GitHub

---

## SECURITY CLEARANCES

**UK SC (Security Check)** - Valid until 2028 **NATO Clearance** - Valid until 2029 **Background Checks:** Watchdog (2024), Sterling (2019)

## AVAILABILITY & CONTACT

**Immediate Availability** for LLM engineering and AI development roles

**Schedule a Discussion:** <https://calendly.com/rifaterdem/schedule> **Email:** [contact@rifaterdemsahin.com](mailto:contact@rifaterdemsahin.com) **Phone:** +44 7848 024173

---

## SUPPORTING DOCUMENTS

**Download Full CV (PDF):** [https://rifaterdemsahin.com/wp-content/uploads/2025/05/erdem-sahin-cv\\_summary\\_2025\\_may.pdf](https://rifaterdemsahin.com/wp-content/uploads/2025/05/erdem-sahin-cv_summary_2025_may.pdf)

**Download Full CV (Word):** [https://rifaterdemsahin.com/wp-content/uploads/2025/05/erdem-sahin-cv\\_summary\\_2025\\_may.docx](https://rifaterdemsahin.com/wp-content/uploads/2025/05/erdem-sahin-cv_summary_2025_may.docx)

**Technical Portfolio & Presentations:** <https://rifaterdemsahin.com/wp-content/uploads/2025/02/rifaterdemsahinprofilepresentation.v2025.2.pdf>

---

*References and detailed project portfolios available upon request*

---

## KEY ACCOMPLISHMENTS

### 2024 | Goldman Sachs | Muscat, Oman

**Enterprise Llm Engineer Platform - Challenge:** Implement scalable llm engineer solutions for financial services environment - **Solution:** Architected comprehensive platform with automation and monitoring capabilities - **Impact:** - 300% improvement in operational efficiency through advanced automation - 30% reduction in operational costs via intelligent optimization - 99.9% system availability with automated failover and recovery - **Technologies:** Kubernetes, Terraform, Azure/AWS, Monitoring tools, Automation frameworks

### 2023 | Ypsomed | Switzerland

**Healthcare Technology Platform - Challenge:** Scale llm engineer capabilities for medical device and IoT environments - **Solution:** Implemented robust platform with real-time processing and analytics - **Impact:** - 40% improvement in system performance through optimized architecture - 35% reduction in operational overhead via automation - Real-time insights enabling improved healthcare delivery - **Technologies:** Cloud platforms, Monitoring systems, Automation tools, Analytics platforms

**2022 | Cushman & Wakefield | London, UK**

**Real Estate Technology Platform - Challenge:** Modernize llm engineer infrastructure for property management analytics - **Solution:** Built cloud-native platform with automated processes and real-time analytics - **Impact:** - 50% faster processing through optimized platform architecture - 90% improvement in system reliability via automated monitoring - Enhanced business intelligence enabling strategic decision making - **Technologies:** Azure services, Analytics platforms, Automation frameworks, Monitoring tools

---

## TECHNICAL EXPERTISE

### Core Technologies

**Platforms:** Kubernetes, Docker, AWS, Azure, GCP **Automation:** Terraform, Ansible, GitLab CI/CD, GitHub Actions **Monitoring:** Prometheus, Grafana, ELK Stack, Azure Monitor **Programming:** Python, Bash, PowerShell, YAML, JSON

### Specialized Skills

**Infrastructure:** Infrastructure as Code, Container orchestration, Cloud architecture **Security:** Zero-trust architectures, Compliance frameworks, Identity management **Operations:** CI/CD pipelines, Monitoring and alerting, Incident response **Integration:** API design, Microservices, Event-driven architectures

---

## PROFESSIONAL EXPERIENCE HIGHLIGHTS

**Senior Llm Engineer / Technical Lead** | 2020 - Present *Goldman Sachs, Ypsomed, Cushman & Wakefield*

- Led llm engineer initiatives across finance, healthcare, and real estate sectors
- Designed and implemented solutions supporting millions of daily transactions
- Established best practices and automated frameworks
- Mentored engineering teams on modern technologies and practices
- Delivered solutions achieving 300% improvement in operational efficiency

**Llm Engineer** | 2016 - 2020 *Microsoft, Emerson, Various Fortune 500*

- Built enterprise solutions for digital transformation initiatives
- Led cross-functional teams designing scalable applications
- Established frameworks and implementation strategies
- Evangelized modern technologies through technical leadership

---

## EDUCATION

**Bachelor of Science** Southern New Hampshire University, USA | 2013

---

## CERTIFICATIONS & CONTINUOUS LEARNING

**Microsoft Certified Solutions Architect Expert AWS Certified Solutions Architect Professional Azure Solutions Architect Expert Certified Kubernetes Administrator (CKA)**

**Continuous Learning:** - Active contributor to open-source projects - Regular participant in technical communities - Following latest developments in technology - Experimenting with emerging tools and practices

---

## SECURITY CLEARANCES

**UK SC (Security Check)** - Valid until 2028 **NATO Clearance** - Valid until 2029 **Background Checks:** Watchdog (2024), Sterling (2019)

---

## AVAILABILITY & CONTACT

**Immediate Availability** for llm engineer roles

**Schedule a Discussion:** <https://calendly.com/rifaterdem/schedule> **Email:** [contact@rifaterdemsahin.com](mailto:contact@rifaterdemsahin.com) **Phone:** +44 7848 024173

---

## SUPPORTING DOCUMENTS

**Download Full CV (PDF):** [https://rifaterdemsahin.com/wp-content/uploads/2025/05/erdem-sahin-cv\\_summary\\_2025\\_may.pdf](https://rifaterdemsahin.com/wp-content/uploads/2025/05/erdem-sahin-cv_summary_2025_may.pdf)

**Download Full CV (Word):** [https://rifaterdemsahin.com/wp-content/uploads/2025/05/erdem-sahin-cv\\_summary\\_2025\\_may.docx](https://rifaterdemsahin.com/wp-content/uploads/2025/05/erdem-sahin-cv_summary_2025_may.docx)

**Technical Portfolio & Presentations:** <https://rifaterdemsahin.com/wp-content/uploads/2025/02/rifaterdemsahinprofilepresentation.v2025.2.pdf>

---

*References and detailed project portfolios available upon request*